STAYING SAFE ON PUBLIC WI-FI

A practical cybersecurity guide for non-technical users

Purpose: This guide explains how everyday internet safety technologies work and gives practical steps for using public networks more safely.

1. Understand the basics

DNS (Domain Name System): DNS helps your device find the network address associated with a domain name such as example.com. Think of it like looking up a business name in an address book. On an untrusted network, attackers may try to manipulate DNS responses or settings and direct users to a different destination. Check website names carefully and do not ignore browser warnings.

HTTPS and TLS: HTTPS is the secure version of HTTP. It commonly uses TLS (Transport Layer Security) to encrypt information travelling between your browser and a website and to help verify the website's identity. This makes it harder for people on the same Wi-Fi to read or alter that traffic. However, a scam website can also use HTTPS, so check the domain name and do not treat the padlock as proof that a site is trustworthy.

VPN (Virtual Private Network): A VPN routes traffic through a VPN server and usually encrypts traffic between your device and that server. It can add protection on public Wi-Fi, but it does not make every website safe, stop every phishing attempt, or remove malware. Choose a reputable provider and avoid unknown VPN apps.

Firewall: A firewall applies rules to network traffic and allows or blocks connections. It can help reduce unwanted network access to your device. Keep your device's firewall enabled and do not switch it off just because a public Wi-Fi sign-in page asks you to.

2. What can go wrong on public Wi-Fi?

Public Wi-Fi is available in cafés, hotels, airports, libraries and other shared places. Risks can include:

- **Fake hotspot:** a network name is made to look like the venue's official Wi-Fi.
- **Eavesdropping:** poorly protected traffic may be intercepted.
- **Evil twin:** a fake access point imitates a legitimate network.
- **Phishing:** a fake page tricks users into revealing passwords or personal information.
- **Unsafe downloads or device sharing:** unexpected downloads or exposed sharing services may put a device at risk.

No single tool removes every risk. HTTPS, software updates, careful browsing and sensible network settings work together to improve safety.

3. Connect safely: a practical checklist

- **Confirm the network name.** Ask venue staff for the exact Wi-Fi name; do not rely on a similar-looking name.
- **Use trusted connections.** For sensitive tasks such as banking, consider using mobile data or another trusted network if available.
- **Check the website address.** Look for the correct domain and HTTPS. Remember: HTTPS does not prove that the website itself is honest.
- **Keep your device updated.** Install operating-system and browser updates from official settings.
- **Use strong, unique passwords.** A password manager and multi-factor authentication can reduce account risks.
- **Turn off unnecessary sharing.** Use your device's public-network setting where available, and disable file/printer sharing when it is not needed.
- **Use a reputable VPN if appropriate.** Install it from an official source and understand that it does not replace HTTPS or safe browsing.
- **Forget the network afterwards.** Turn off Wi-Fi when you no longer need it and remove saved networks you do not trust.
- **Avoid unexpected prompts.** Do not install unknown apps, certificates, profiles or browser extensions to get Wi-Fi access.

4. Recognise suspicious captive portals

A **captive portal** is a sign-in or acceptance page shown before a public Wi-Fi network provides internet access. Many are legitimate, but criminals can imitate them. Treat these signs as reasons to stop and verify:

- The page asks for bank details, card PINs, one-time passcodes (OTPs), or passwords for unrelated accounts.
- The website address is misspelled, unrelated to the venue, or unexpectedly changes to an unfamiliar domain.
- The page pressures you with threats or extreme urgency.
- It asks you to download an unknown app, install a certificate/profile, or bypass a browser security warning.
- The venue's staff cannot confirm that the page or network is official.

What to do: Stop entering information. Disconnect from the network, verify the Wi-Fi name and portal with venue staff using a trusted contact method, and report suspicious pages. Never bypass a certificate warning just to connect.

5. Quick safety checklist

Before connecting	While browsing	When finished
■ Confirm the exact network name. ■ Keep device and browser updated. ■ Use a trusted connection for sensitive work.	■ Check domain names. ■ Avoid suspicious portal requests. ■ Keep firewall enabled; use VPN if appropriate.	■ Sign out of accounts on shared devices. ■ Disconnect and forget untrusted networks. ■ Report suspicious pages.

6. Conclusion

Public Wi-Fi can be useful when used carefully. Understand the roles of DNS, HTTPS/TLS, VPNs and firewalls; verify networks and website addresses; keep devices updated; and treat unexpected login or installation requests with caution. When uncertain, disconnect and ask the venue to verify the official network.

References

- Cloudflare, "What is DNS?" — https://www.cloudflare.com/learning/dns/what-is-dns/
- Cloudflare, "What is HTTPS?" — https://www.cloudflare.com/learning/ssl/what-is-https/
- Cybersecurity and Infrastructure Security Agency (CISA) — https://www.cisa.gov/
- Federal Trade Commission (FTC), Consumer Advice — https://consumer.ftc.gov/
- Microsoft Support — https://support.microsoft.com/

Prepared as general cybersecurity awareness material. Advice may vary by device, venue and service; verify details with official documentation.